Odd Radicals and Products of Six Linear Factors

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Abstract

For a fixed product of six rational linear factors, we exhibit three obstructions to bounding a multiplicative scale by the odd radical of the product at two arguments. A rational involution gives a negative square for every prescribed rational scale greater than four. A second involution gives positive examples of unbounded scale whose first argument is a sum of two rational squares. A partial symmetry yields an elliptic family with both arguments positive and the first greater than one. We determine all rational Möbius transformations preserving the six factors and give exact certificates for the identities and the elliptic non-torsion calculation. The results concern height bounds for the product; they do not assert solutions of the auxiliary elliptic and pole equations in a Diophantine definition.

1 Definitions and the height question

For a nonzero rational number q, let $v_p(q)$ be its normalized p-adic valuation. For a finite set S of rational primes, put

$$\mathrm{Odd}_S(q) = \prod_{\substack{p \notin S \\ v_p(q) \text{ odd}}} p.$$

Negative valuations are included in this definition. In particular $\mathrm{Odd}_S(cq^2) = \mathrm{Odd}_S(c)$ for $c, q \in \mathbb{Q}^\times$. If $x = u/v$ in lowest terms with $v > 0$, write $h(x) = \log \max(|u|, v)$.

Set

$$\Theta = \left\{0, -\frac{1}{2}, -\frac{1}{4}, -\frac{2}{5}, -1, -\frac{1}{3}\right\}, \tag{1}$$

$$F(t) = \prod_{\theta \in \Theta} (t - \theta) = \frac{t(t+1)(2t+1)(3t+1)(4t+1)(5t+2)}{120}, \tag{2}$$

$$R(t) = \frac{F(t)}{(t-1)^8}. \tag{3}$$

An admissible pair (a, λ) consists of nonzero rational numbers such that both λ and $a\lambda$ lie outside $\Theta \cup \{1\}$. We examine bounds of the form

$$h(a) \le C \log \mathrm{Odd}_S(R(\lambda)R(a\lambda)) + C' \tag{4}$$

with constants independent of the admissible pair. The denominator in (3) is a square, but we retain it to record all poles and to match the rational functions used in the six-contact construction of [1].

Theorems 2.2, 3.2 and 4.2 give families for which the radical in (4) is constant and $h(a)$ is unbounded. Their domains differ: the first permits every prescribed scale $a > 4$, the second imposes a Gaussian norm condition on λ, and the third imposes $\lambda > 1$. The additional conditions are stated separately throughout.

2 Cancellation for every prescribed scale

Define

$$\psi(t) = -\frac{t}{5t+1}.$$

This involution fixes $0, -2/5$ and exchanges $-1/2$ with $-1/3$ and $-1/4$ with -1.

Lemma 2.1. *As identities of rational functions,*

$$F(\psi(t)) = -\frac{F(t)}{(5t+1)^6}, \tag{5}$$

$$R(t)R(\psi(t)) = -\left(\frac{F(t)(5t+1)}{(t-1)^4(6t+1)^4}\right)^2. \tag{6}$$

Proof. Put $N = -t$ and $D = 5t+1$. The five remaining numerator factors in $120F(N/D)$ are

$$N + D = 4t + 1, \quad 2N + D = 3t + 1, \quad 3N + D = 2t + 1,$$

$$4N + D = t + 1, \qquad 5N + 2D = 5t + 2.$$

Their product with N is $-120F(t)$, which proves (5). Since $\psi(t) - 1 = -(6t+1)/(5t+1)$, substitution into (3) gives (6). □

Theorem 2.2. *For every rational $a > 4$, set*

$$\lambda = -\frac{a+1}{5a}.$$

Then (a, λ) is admissible, $a\lambda = \psi(\lambda)$, and

$$R(\lambda)R(a\lambda) = -\left(\frac{25a(a-4)(a-1)(a+1)(2a-3)(3a-2)(4a-1)}{24(a+6)^4(6a+1)^4}\right)^2. \tag{7}$$

Every finite valuation of this nonzero product is even. Thus $\mathrm{Odd}_S(R(\lambda)R(a\lambda)) = 1$ for every finite set S.

Proof. One has

$$5\lambda + 1 = -1/a, \qquad \psi(\lambda) = -(a+1)/5 = a\lambda.$$

The inequalities

$$-1/4 < \lambda < -1/5, \qquad a\lambda < -1$$

exclude all six contacts and the pole 1. Moreover

$$\lambda - 1 = -\frac{6a+1}{5a}, \qquad 6\lambda + 1 = -\frac{a+6}{5a},$$

so the transformed denominators do not vanish. Lemma 2.1 proves the negative-square assertion. Substitution and cancellation yield (7), whose displayed factors are nonzero for $a > 4$. □

Corollary 2.3. *Fix rational numbers $B \geq 3$ and $T > 0$. The inequality (4) cannot hold for every admissible pair with*

$$a = (2X^2 + B)^8 + T^8, \qquad X \in \mathbb{Q},$$

for any constants C, C' and any fixed finite set S.

Proof. Each displayed scale exceeds four, so Theorem 2.2 gives an admissible slope with product odd radical one. Taking positive integers X tending to infinity makes a and hence $h(a)$ tend to infinity. □

In particular the restriction $a = \beta^8 + T(n)^8$ in the weak-*abc* construction does not, by itself, prevent cancellation: for a fixed nonzero elliptic index n, its trace $T(n)$ is a fixed positive rational number and $\beta = 2X^2 + B$. The product in (7) is negative. The theorem asserts neither the auxiliary local tests nor solvability of a pole predicate.

3 Contact automorphisms and positive norm slopes

Proposition 3.1. *There are exactly twelve rational Möbius transformations preserving* Θ. *They are listed below. A row* $(a, b, c, d; k)$ *denotes*

$$\gamma(t) = \frac{at + b}{ct + d}, \qquad F(\gamma(t))(ct + d)^6 = kF(t).$$

a	b	c	d	k
1	0	-5	-1	-1
1	0	0	1	1
1	1	-7	-4	-27
1	1	2	-1	27
2	1	-7	-3	1
2	1	-3	-2	-1
3	1	-8	-3	-1
3	1	-7	-2	1
4	1	-13	-4	27
4	1	-7	-1	-27
5	2	-14	-5	-27
5	2	-11	-5	27

Apart from the identity, the only transformation sending some positive rational number to a positive number is

$$\tau(t) = \frac{t + 1}{2t - 1}.$$

Its positive domain is $t > 1/2$. *For* $t > 1$ *it satisfies* $0 < \tau(t)/t < 2$.

Proof. A Möbius transformation is determined by its values at the three distinct contacts $0, -1/2, -1/4$. There are exactly $6 \cdot 5 \cdot 4 = 120$ possible ordered triples of distinct images in Θ. For a source triple (x_i) and target triple (y_i), its coefficient vector spans the kernel of

$$\begin{pmatrix} x_1 & 1 & -x_1y_1 & -y_1 \\ x_2 & 1 & -x_2y_2 & -y_2 \\ x_3 & 1 & -x_3y_3 & -y_3 \end{pmatrix}.$$

The signed 3×3 minors give that vector explicitly. Testing the remaining three contacts leaves exactly the twelve rows of the table. This is a finite rational calculation; the accompanying checker performs all 120 cases and verifies each polynomial identity.

In each row other than the identity and τ, the numerator coefficients are positive or zero and the denominator coefficients are negative. Such a transformation sends every $t > 0$ to

a negative number. The asserted positive domain of τ follows directly from its denominator. Finally, for $t > 1$, the inequality $\tau(t)/t < 2$ is equivalent to

$$4t^2 - 3t - 1 = (t-1)(4t+1) > 0.$$

□

The rational function identity for τ in the table gives

$$R(t)R(\tau(t)) = 3\left(\frac{3F(t)(2t-1)}{(t-1)^4(t-2)^4}\right)^2. \tag{8}$$

Here $\tau(t) - 1 = (2-t)/(2t-1)$ explains the additional denominator.

Theorem 3.2. *For every integer $m \geq 2$, set*

$$\lambda_m = \frac{m^2+1}{2m^2}, \qquad \mu_m = \frac{3m^2+1}{2}, \qquad a_m = \frac{m^2(3m^2+1)}{m^2+1}.$$

Then (a_m, λ_m) is admissible, $a_m > 4$, $\mu_m = a_m\lambda_m = \tau(\lambda_m)$, and $a_m \to \infty$. Both arguments are positive and

$$\lambda_m = \left(\frac{m+1}{2m}\right)^2 + \left(\frac{m-1}{2m}\right)^2. \tag{9}$$

Moreover $R(\lambda_m)R(a_m\lambda_m) \in 3\mathbb{Q}^{\times 2}$. Thus (4) fails even with positive Gaussian norm slopes when $3 \in S$.

Proof. All relations are immediate substitutions. In particular

$$1/2 < \lambda_m \leq 5/8, \qquad \mu_m \geq 13/2,$$

so no argument is a contact or pole. Also $a_m = 3m^2 - 2 + 2/(m^2+1) > 4$ and $a_m \to \infty$. Equation (9) proves the norm condition, and (8) proves the product squareclass assertion. □

For a prescribed rational scale $a > 0$, the equation $\tau(\lambda) = a\lambda$ is

$$2a\lambda^2 - (a+1)\lambda - 1 = 0.$$

It has a rational root precisely when $a^2 + 10a + 1$ is a rational square. Theorem 3.2 does not impose the further condition $a = \beta^8 + T(n)^8$. Also, Proposition 3.1 shows that no nonidentity contact automorphism gives $\gamma(\lambda) = a\lambda$ with $\lambda > 1$ and $a > 4$. That observation concerns contact automorphisms only; the next section gives an obstruction of a different kind.

4 An elliptic family above one

Put

$$\gamma(t) = \frac{1}{2(t-1)}, \qquad H(t) = (2t-1)(3t+1)(5t+2).$$

Direct multiplication gives

$$F(t)F(\gamma(t)) = \left(\frac{t(t+1)(2t+1)(4t+1)}{480(t-1)^3}\right)^2 H(t). \tag{10}$$

Thus a rational point on $7y^2 = H(t)$, away from the excluded arguments, gives $R(t)R(\gamma(t)) \in 7\mathbb{Q}^{\times 2}$.

Lemma 4.1. *The elliptic curve*

$$\mathcal{E}: \quad Y^2 = (X-105)(X+70)(X+84)$$

contains the non-torsion point $P = (210, 2940)$. *This point belongs to the identity component of* $\mathcal{E}(\mathbb{R})$.

Proof. Substitution verifies that P lies on the curve. The three roots are distinct modulo 11 and modulo 13, so both primes are of good reduction. Exact enumeration and the group law give

$$\#\mathcal{E}(\mathbb{F}_{11}) = 8, \qquad \operatorname{ord}(P \bmod 13) = 10.$$

For clarity the successive nonidentity multiples modulo 13 are

$$(2,2), (11,4), (0,10), (4,6), (1,0), (4,7), (0,3), (11,9), (2,11).$$

The next multiple is the identity. If P were torsion, its order would therefore be divisible by five. Rational torsion of order prime to a good reduction prime injects under reduction [3, Chapter VII, Section 3]. At 11 this would give an element of order five in a group of order eight, a contradiction.

The largest real root of the cubic is 105, and $X(P) = 210 > 105$. Hence P is on the unbounded real branch, which contains the point at infinity and is the identity component. □

Theorem 4.2. *There are rational sequences* $\lambda_j > 1$ *and* $a_j > 4$ *with* $a_j \to \infty$ *such that* (a_j, λ_j) *is admissible and*

$$R(\lambda_j)R(a_j\lambda_j) \in 7\mathbb{Q}^{\times 2}.$$

Consequently (4) *fails on the domain* $\lambda > 1$, $a > 4$, *whenever* $7 \in S$.

Proof. The change of variables $X = 210t$, $Y = 1470y$ identifies $7y^2 = H(t)$ with $\mathcal{E}$. Indeed

$$(210t-105)(210t+70)(210t+84) = 308700H(t), \qquad 1470^2 = 7 \cdot 308700.$$

The point P corresponds to $(t, y) = (1, 2)$.

The identity component of the real points of an elliptic curve is a circle group. The multiples of a non-torsion point are dense in that component: after identification with $\mathbb{R}/\mathbb{Z}$, such a point has irrational coordinate, and its cyclic subgroup is dense. By Lemma 4.1, choose rational multiples P_j tending to P from the local side where $X > 210$. This side exists because $Y(P) \neq 0$, so X is a local real coordinate at P. Then

$$\lambda_j = X(P_j)/210 > 1, \qquad \lambda_j \longrightarrow 1.$$

Set

$$\mu_j = \frac{1}{2(\lambda_j - 1)}, \qquad a_j = \frac{\mu_j}{\lambda_j} = \frac{1}{2\lambda_j(\lambda_j - 1)}.$$

For all sufficiently large j, $\mu_j > 4\lambda_j > 4$ and the arguments avoid every contact and pole. Equation (10) proves the squareclass assertion. Finally $a_j \to \infty$, and therefore $h(a_j) \to \infty$. □

No assertion in Theorem 4.2 imposes that $\lambda_j - 1$ be a sum of two rational squares or that $a_j = \beta^8 + T(n)^8$. Adding either condition requires a further argument. In particular, this theorem does not settle the product height question on their intersection.

5 Quadratic twists and exact verification

The geometric meaning of the cancellation is straightforward. For a squarefree polynomial f, the equation

$$z^2 = f(x)f(y)$$

records pairs of points on the same quadratic twist of $w^2 = f(x)$. Allowing a fixed nonzero factor on the right records pairs of twists with a fixed ratio of squareclasses. This pair surface, including its rational curves and its relation to uniform height questions, is studied in [2, Sections 7–8]. Finiteness of rational points on each fixed genus-two twist does not give a bound uniform in a varying squareclass. Consequently, excluding contact automorphism families alone does not establish (4).

All identities and finite computations in this paper are verified by `check_product_height_obstructions_v19.py` in `computation/pole_parameters`. The program uses only rational arithmetic and the Python standard library. It checks the two homogeneous sextic identities, the explicit square root in (7), the complete 120-case automorphism enumeration, and the cleared polynomial identity (10). It also verifies the cubic change of variables, enumerates the groups over $\mathbb{F}_{11}$ and $\mathbb{F}_{13}$, and checks the stated point orders. Its certificate records these finite results. The argument giving the infinite elliptic sequence is the real-density proof of Theorem 4.2.

References

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